

Function Object aur NFE

Functions objects hote hain

JS me function ek **object** hai jise call bhi kar sakte ho. Isliye usme properties add/remove kar sakte ho.

Built-in properties

name

Function ka naam. Agar naam nahi diya aur variable me assign kiya, to JS variable ke naam se guess kar leta hai.

javascript

```
let sayHi = function () {};  
sayHi.name; // "sayHi"
```

Array ke andar bane function ka naam khaali hota hai, kyunki guess karne ka context nahi milta.

length

Function ke **parameters ki ginti**. Rest parameters (**...more**) count nahi hote.

javascript

```
function f(a, b, ...rest) {}  
f.length; // 2
```

Custom properties

Function me apni property bhi jod sakte ho.

javascript

```
function sayHi() {  
  sayHi.counter++;  
}  
sayHi.counter = 0;
```

Dhyan: **sayHi.counter** aur function ke andar **let counter** do alag cheezein hain, unka aapas me koi link nahi.

Closure vs function property

- **Closure variable:** bahar ka code usse chhu nahi sakta (private).

- **Function property:** bahar se change ho sakta hai (`counter.count = 10`).

Zarurat ke hisaab se chuno.

Named Function Expression (NFE)

Function expression ko naam dena:

javascript

```
let sayHi = function func(who) {  
  if (who) console.log("Hello " + who);  
  else func("Guest"); // khud ko call kiya  
};
```

Do khaas baatein:

1. Function is naam se **andar se khud ko refer** kar sakta hai.
2. Ye naam **bahar se dikhta nahi**.

Ye kaam kyun aata hai? Agar andar (`sayHi(...)`) likhte aur baad me bahar (`sayHi = null`) ho jata, to function toot jata. (`func`) naam hamesha usi function ko point karta hai.

Ye trick **Function Declaration** me nahi chalti.

Libraries kaise use karti hain

jQuery ka (\$) aur lodash ka (_) ek function hain, jinke upar bahut saari helper functions properties ki tarah lagi hoti hain. Isse sirf ek global naam use hota hai.

Yaad rakho

- Function = callable object.
- (`name`) aur (`length`) built-in properties hain.
- Properties aur variables alag hote hain.
- NFE se function safely khud ko call kar sakta hai.